

Simulated Phase I Trial EDC Design

REDCap Project Demonstration

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Project Overview

- **Project Type:** Simulated EDC Design (Academic/Portfolio Project)
- **Tool Used:** REDCap
- **Study Focus:** Simulated first-in-human vaccine study
- **Goal:** Build a working EDC prototype mimicking a vaccine Phase I clinical trial

Trial Design

- **Arms:** 12 participants in each group (high, medium and low dose) including 3 placebo participants.
- **Schedule:** Screening (W-1), Dosing (W0), Boosters (W3, W6), Follow-up (W1–W16).
- **Visits:** 10 total visits.
- **Design Considerations:**
 - Equal visit windows across arms.
 - Designed for clean source-to-CRF alignment.

Visit Schedule

- Defined 3 study arms with shared visit structure.
- Set up scheduled visits (e.g., Screening, Baseline, Follow-up, EOS).

Arm 1: Arm 1 Arm 2: Arm 2 Arm 3: Arm 3 +Add New Arm

Arm name: Arm 1 [Rename Arm 1](#) | [Delete Arm 1](#)

		Event # [event-number]	Event Label [event-label]	Custom Event Label [optional]	Unique event name (auto-generated) [event-name]	Event ID (auto-generated, unchangeable) [event-id]
		1	Screening	Week -1	screening_arm_1	106350
		2	Baseline	Week 0, dose 1	baseline_arm_1	106351
		3	Followup 1	Week 1	followup_1_arm_1	106352
		4	Followup 2	Week 4, booster 1	followup_2_arm_1	106354
		5	Followup 3	Week 5	followup_3_arm_1	106355
		6	Followup 4	Week 8, booster 2	followup_4_arm_1	106358
		7	Followup 5	Week 9	followup_5_arm_1	106359
		8	Followup 6	Week 12	followup_6_arm_1	106360
		9	Followup 7	Week 14	followup_7_arm_1	106361
		10	EOS	Week 16	eos_arm_1	106362

Event Mapping

- Mapped each instrument (CRF) to relevant events.
- Demonstrates understanding of longitudinal study design and CRF event alignment.

Arm 1: Arm 1 Arm 2: Arm 2 Arm 3: Arm 3										
Arm name: Arm 1										
Begin Editing Save										
Data Collection Instrument	Screening (1)	Baseline (2)	Followup 1 (3)	Followup 2 (4)	Followup 3 (5)	Followup 4 (6)	Followup 5 (7)	Followup 6 (8)	Followup 7 (9)	EOS (10)
Participant Entry - Screening Survey	✓									
Clinical Staff Entry - Screening	✓									
Clinical Staff Entry - Baseline Visit		✓								
Clinical Staff Entry - Follow-up Form			✓	✓	✓	✓	✓	✓	✓	✓
Clinical Staff Entry - Dose Administration Form		✓		✓		✓				
Clinical Staff Entry - Sample Collection Form		✓	✓	✓	✓	✓	✓	✓	✓	✓
NCI Adverse Event Serious Adverse Event CTCAE V4 3265657v3 0		✓	✓	✓	✓	✓	✓	✓	✓	
Clinical Staff Entry - Final Safety Review										✓

CRF Instruments

Instrument name	Fields	View PDF
Participant Entry - Screening Survey	29	
Clinical Staff Entry - Screening	23	
Clinical Staff Entry - Baseline Visit	28	
Clinical Staff Entry - Follow-up Form	27	
Clinical Staff Entry - Dose Administration Form	8	
Clinical Staff Entry - Sample Collection Form	8	
NCI Adverse Event Serious Adverse Event CTCAE V4 3265657v3 0	26	
Clinical Staff Entry - Final Safety Review	8	

- **Participant Screening Surveys:** Demographics, Medical History, Lifestyle.
- **Clinical Staff Entries:** Screening , Baseline Vital Signs, Dose Administration, Sample Collection Form, AE/SAE Forms.

Custom Instruments Created	Purpose & Key Features
Participant Entry – Screening Survey	Self-reported demographics, lifestyle, diet, allergies, and eligibility questions.
Clinical Staff Entry – Screening	Vital signs, pregnancy check, exclusion criteria, and initial condition assessment.
Clinical Staff Entry – Baseline Visit Clinical Staff Entry – Follow-up Visit	Physical exam, medical history update, and ongoing condition review.
Dose Administration Form	Captures planned dose group, product info, lot number, and administration details.
Sample Collection Form	Tracks blood and urine samples by visit, including collection date and sample types.
AE/SAE Form (import from library)	Based on NCI CTCAE v4 for adverse event and serious adverse event documentation.
Final Safety Review Form	Summarizes participant's safety data after study completion.

CRF Instruments

Participant Entry – Screening Survey

Informed Consent

Add Field Add Matrix of Fields Import from Field Bank

Field Name: consent_form

Consent Form

Attachment: Consent-Form-Template.pdf (97.2 kB)

Add Field Add Matrix of Fields Import from Field Bank

Field Name: signed_consent

Signed / Scanned Form

* must provide value [Upload file](#)

Add Field Add Matrix of Fields Import from Field Bank

Demographics

Add Field Add Matrix of Fields Import from Field Bank

Field Name: sex

What is your sex assigned at birth?

* must provide value

Add Field Add Matrix of Fields Import from Field Bank

Field Name: pregnancy Branching logic: [sex] = '2'

Are you currently pregnant or planning to become pregnant in the near future?

* must provide value

☐ Yes, I am currently pregnant.
☐ No, I am not pregnant and do not plan to become pregnant in the near future.
☐ I am unsure or prefer not to answer.

reset

Key Feature Highlight

- Informed Consent
 - Enabled **file upload** for participants to submit signed eConsent forms.
 - Designed to reflect common clinical-trial informed-consent documentation considerations.
- Demographics with Conditional Logic
 - If "Female" is selected in the sex question, a **Pregnancy Status** question appears.
 - Shows ability to implement **branching logic** for adaptive surveys.

CRF Instruments

Clinical Staff Entry – Screening

Field Name: `weight_screen` Branching logic: `[vital_sign] = 'Y'`

Weight (kg)

* must provide value

Add Field Add Matrix of Fields Import from Field Bank

Field Name: `height` Branching logic: `[vital_sign] = 'Y'`

Height (cm)

* must provide value

Add Field Add Matrix of Fields Import from Field Bank

Field Name: `bmi_screen` Branching logic: `[vital_sign] = 'Y'`

BMI

* must provide value View equation

Clinical Staff Entry – Baseline Visit

Vaccination

Add Field Add Matrix of Fields Import from Field Bank

Field Name: `randomized_group`

Randomized Group

* must provide value

☐ Arm 1 - Active
☐ Arm 1 - Placebo
☐ Arm 2 - Active
☐ Arm 2 - Placebo
☐ Arm 3 - Active
☐ Arm 3 - Placebo

reset

Key Feature Highlight

- Vital Signs
 - Implemented a calculated field for automatic BMI computation.
 - Demonstrates understanding of clinical data elements and use of REDCap calculated fields.
- Randomized Group
 - Randomized the arm of the participant.
 - Demonstrates understanding of REDCap randomization features and their application in clinical study design.

Note: REDCap test instance does not support production features like randomization setup. Randomization field included as structural placeholder.

CRF Instruments

Clinical Staff Entry – Sample Collection Form

Field Name: sample_type

Sample Type

* must provide value

☐ Whole Blood
☐ Serum
☐ Plasma
☐ PBMCs
☐ Other

Please follow the format below for each sample collected:

SIM-001XXXWBVYAA

XXX = Subject number (3 digits, e.g., 001)

WB = Sample type (e.g., WB = Whole Blood, P = Plasma, S = Serum, PBMCs=PBMC)

VY = Visit number (e.g., V1, V2...)

AA = Aliquot number (e.g., 01, 02...)

Field Name: blood_sample_id Branching logic: [sample_type[1]] = '1'

Whole Blood Sample ID

* must provide value

@CHARLIMIT

Field Name: serum_sample_id Branching logic: [sample_type[2]] = '1'

Key Feature Highlight

- Sample ID with Manual Format Enforcement
 - Created structured instructions for Sample ID entry (e.g., SIM-001-001WBV1-01).
 - Used @CHARLIMIT=20 to reduce errors despite the lack of regex module support.

Clinical Sample Collection Tracking

- **Purpose:**
 - Ensures traceability and reliable linkage to clinical data.
 - Reflects awareness of lab traceability requirements from prior hands-on sample testing experience.
- **Custom Sample ID format:** designed using lab-informed structure for clarity and consistency.
- **Field control:** implemented field instructions and @CHARLIMIT to enforce format without external modules.

SIM-001-XXXWBVYAA

XXX = Subject number (3 digits, e.g., 001)

WB = Sample type (e.g., WB = Whole Blood, P = Plasma, S = Serum, PBMCs=PBMC)

VY = Visit number (e.g., V1, V2...)

AA = Aliquot number (e.g., 01, 02...)

AE/SAE Tracking & Final Safety Review

- Imported **NCI AE/SAE form** to document severity, relatedness, and outcomes.
- Created Final Safety Review to summarize subject safety at study completion.
- Designed to reflect common clinical-trial safety data documentation concepts.

Current instrument: **NCI Adverse Event Serious Adverse Event CTCAE V4 3265657v3 0** [Preview instrument](#)

The screenshot shows a form titled "NCI Adverse Event Serious Adverse Event CTCAE V4 3265657v3 0". It has a header bar with "Add Field", "Add Matrix of Fields", and "Import from Field Bank" buttons. Below the header is a section for "Adverse Event Grade" with a "Field Name: cadsr_2944515_y1_0" and a "reset" button. The "Adverse Event Grade" section contains a list of radio buttons: "Mild Adverse Event", "Moderate Adverse Event", "Severe Adverse Event", "Life Threatening Adverse Event", and "Death Related to Adverse Event". Below this list is a note: "A coded value to identify the severity of the AE." Below the "Adverse Event Grade" section is a section for "Conditional Adverse Event Questions" with a "Field Name: cadsr_2744993_y1_0" and a "reset" button. The "Conditional Adverse Event Questions" section contains a list of radio buttons: "Unknown". Below this list is a note: "Date on which the AE was first evident. Conditionality Rule: Condition is that this is based on the needs of the Sponsor. CTCP does not currently collect Adverse Event onset dates for routine AEs reported via reporting systems."

AE/SAE form

Current instrument: **Clinical Staff Entry - Final Safety Review** [Preview instrument](#)

The screenshot shows a form titled "Clinical Staff Entry - Final Safety Review". It has a header bar with "Add Field", "Add Matrix of Fields", and "Import from Field Bank" buttons. Below the header is a section for "Final Safety Review" with a "Field Name: final_ae" and a "reset" button. The "Final Safety Review" section contains a list of radio buttons: "No" and "Yes". Below this list is a note: "* must provide value". Below the "Final Safety Review" section is a section for "Did the subject experience any serious adverse events (SAEs)?" with a "Field Name: final_sae" and a "reset" button. The "Did the subject experience any serious adverse events (SAEs)?" section contains a list of radio buttons: "No" and "Yes". Below this list is a note: "* must provide value".

Final Safety Review

Randomization Setup

Demo Only – Test Instance Limitation

The screenshot shows the 'Vaccination' form in REDCap. At the top, there are buttons for 'Add Field', 'Add Matrix of Fields', and 'Import from Field Bank'. Below these, the 'Field Name' is set to 'randomized_group'. The 'Randomized Group' section contains a list of radio button options: 'Arm 1 - Active', 'Arm 1 - Placebo', 'Arm 2 - Active', 'Arm 2 - Placebo', 'Arm 3 - Active', and 'Arm 3 - Placebo'. A red asterisk indicates that a value must be provided. A 'reset' button is located at the bottom right of the form.

- Step 1: Move project to production and enable randomization.
- Step 2: Set up Randomization.
 - Define `randomized_group` as randomization field, use `gender` as stratification variable.
 - Create and upload an allocation table (CSV) with a 9:3 ratio of active:placebo per arm, stratified by gender.
- Step 3: REDCap will automatically assign a group when a record is randomized.

1	stratum,treatment_group	
2	Male,A1_M	
3	Male,A1_M	
4	Male,A1_M	
5	Male,A1_M	
6	Male,A1_M	
7	Male,A1P_M	
8	Female,A1_F	
9	Female,A1_F	
10	Female,A1_F	
11	Female,A1_F	
12	Female,A1P_F	
13	Female,A1P_F	
14	Male,A2_M	
15	Male,A2_M	
16	Male,A2_M	
17	Male,A2_M	
18	Male,A2P_M	
19	Male,A2P_M	
20	Female,A2_F	
21	Female,A2_F	
22	Female,A2_F	

User Rights & Access Control

Role	Account	Typical Access Level
Principal Investigator (PI)	Yes	Read-only or Full Access (if review needed)
Clinical Staff	Yes	View & Edit data; No setup rights
Data Manager	Yes	Full Access; setup tools, reports, data export
CRA / Monitor	Yes	Read-only + report viewing
Statistician / Analyst	Yes	De-identified data export only
Participant	No	Access via Survey Link only

- Assigned role-based access to clinical staff and participants.
- Limited visibility and editing based on responsibilities.
- Reinforces data protection and workflow compliance (HIPAA / GCP-aligned).

Role name (click role name to edit role)	Username or users assigned to a role (click username to edit or assign to role)	Expiration (click expiration date to edit)	Project Design and Setup	User Rights	Data Access Groups	Data Viewing Rights	Data Export Rights	Alerts & Notifications	Reports & Report Builder	Stats & Charts
—	cdcd68192000@gmail.com (YuShuang Sun)	never	✓	✓	✓	8 View & Edit	8 Full Data Set	✓	✓	✓
Clinical Staff	[No users assigned]		✗	👁	✗	8 View & Edit	8 No Access	✗	✗	✗
CRA / Monitor	[No users assigned]		✗	👁	✗	8 View & Edit	8 De-Identified	✗	✗	✗
Data Manager	[No users assigned]		✓	✓	✓	8 View & Edit	8 Full Data Set	✓	✓	✓
PI	[No users assigned]		✓	✓	✓	8 View & Edit	8 Full Data Set	✓	✓	✓
Statistician / Analyst	[No users assigned]		✗	✗	✗	8 Read Only	8 Full Data Set	✗	✗	✓

Clinical Reports

Participants with AE Reported

- Displays unique participants who have reported adverse events (AE) during the study.
- Uses advanced filter logic to include only completed AE forms.
- Helps track overall safety and monitor AE occurrence without duplicate counts.

STEP 3

☐ Show data for all events or repeating instruments/events for each record returned ?

▼ Filters (optional)

Advanced filter logic:
[nci_adverse_event_serious_adverse_event_ctcae_v4_3_complete] = "2"

Participants with AE Reported

Search

Record ID record_id	Event Name redcap_event_name	Complete? nci_adverse_event_serious_adverse_event_ctcae_v4_3_complete
<u>1</u>	Baseline (Arm 1: Arm 1)	Complete (2)
<u>2</u>	Followup 1 (Arm 2: Arm 2)	Complete (2)
<u>2</u>	Baseline (Arm 2: Arm 2)	Complete (2)

Clinical Reports

Sample Collected at Baseline Visit

- Lists participants with clinical samples collected at baseline visit.
- Includes sample type and unique Sample ID for traceability.
- Demonstrates proper linkage between clinical data and lab samples.

STEP 2

Fields to include in report [+ Quick Add](#) Add all fields from selected instrument: ...

Field 1	record_id "Record ID"		Instrument: Participant Entry - Screening S...
Field 2	sample_type "Sample Type"		Instrument: Clinical Staff Entry - Sample Co...
Field 3	blood_sample_id "Whole Blood Sample ID"		Instrument: Clinical Staff Entry - Sample Co...
Field 4	serum_sample_id "Serum Sample ID"		Instrument: Clinical Staff Entry - Sample Co...
Field 5	plasma_sample_id "Plasma Sample ID"		Instrument: Clinical Staff Entry - Sample Co...
Field 6	pbmc_sample_id "PBMCs Sample ID"		Instrument: Clinical Staff Entry - Sample Co...

Sample Collected at Baseline Visit

Search

Record ID record_id	Event Name redcap_event_name	Sample Type					Whole Blood Sample ID blood_sample_id	Serum Sample ID serum_sample_id	Plasma Sample ID plasma_sample_id	PBMCs Sample ID pbmc_sample_id
		Whole Blood sample_type__1	Serum sample_type__2	Plasma sample_type__3	PBMCs sample_type__4	Other sample_type__0				
3	Baseline (Arm 2: Arm 2)	Checked (1)	Checked (1)	Checked (1)	Checked (1)	Unchecked (0)	VXX-401-003WBV101	VXX-401-003SV101	VXX-401-003PV101	VXX-401-003PBMCV101
4	Baseline (Arm 2: Arm 2)	Checked (1)	Checked (1)	Unchecked (0)	Unchecked (0)	Unchecked (0)	VXX-401-004WBV101	VXX-401-004SV101		

Data Quality & Compliance

- **De-identification:** Applied date shifting and avoided PHI fields to align with privacy standards.
- **Field Validation:** Used range checks, required fields, and calculated BMI for data consistency.
- **Compliance Standards:** Incorporated elements from HIPAA, 21 CFR Part 11, and CDASH-style naming for structure and traceability.

De-identification options (optional)

The options below allow you to limit the amount of sensitive information that you are exporting out of the project. Check all that apply.

Known Identifiers:

- ☒ Remove All Identifier Fields (tagged in Data Dictionary)
- ☐ Hash the Record ID field (converts record name to an unrecognizable value)

Free-form text:

- ☐ Remove unvalidated Text fields (i.e. Text fields other than dates, numbers, etc.)
- ☐ Remove Notes/Essay box fields

Date and datetime fields:

- ☐ Remove all date and datetime fields
- OR —
- ☒ Shift all dates by value between 0 and 364 days (shifted amount determined by algorithm for each record)
[What is date shifting?](#)

[Deselect all options](#)

Challenges & Learnings

- **Protocol to CRF:** Translated trial design into structured REDCap forms.
- **No External Modules:** Used action tags (e.g., @CHARLIMIT) to work around system limits.
- **PHI & Data Privacy:** Practiced de-identification (e.g., date shifting) to ensure compliance.
- **User Rights Management:** Set role-based access to support data privacy and workflow needs.
- **Data Quality:** Applied validation, calculations, and formatting to ensure clean data.
- **Logic Building:** Python learning helped sharpen conditional thinking and data logic.